

# Areg Hovumyan

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## Education

California State Polytechnic University, Pomona  
Bachelor of Science in Computer Engineering — GPA: 3.6/4.0

Pomona, CA  
Expected Dec. 2027

## Experience and Projects

### Electrical/Embedded Systems Member: L'SPACE (Mission Concept)

May 2026 – Present

NASA's student acceleration program - Robotic Lunar Rover Mission

Remote

- Presented the Communication and Data Handling to NASA engineers, got feedback, and improved the tools being used by choosing the right software based on their advice.
- Managed to stay within the schedule, cost, and mass limitations (under 250M, 170Kg, no sunlight for 14 days and more) for a possible mission on the moon.
- Designed the whole rover's Communication and Data Handling(C&DH) architecture by communicating with the mechanical and electrical team.
- Established (on concept) the communication between the Earth and the Moon, autonomous mission completion in case of redundancy using NeoPan, learned F<sup>7</sup> to fit the mission under the mission needs.

### Software Engineering Team Lead: Autonomous Hexacopter (University Club)

Jun. 2025 – Present

Lockheed Martin sponsored project

Pomona, CA

- Applied pair programming, test-first development, unit-testing, containerized a ROS2 environment with Docker for a team of 8 student-engineers to collaborate, and built CI/CD pipelines to ship features with fewer defects.
- Collaborated with the Electrical team to not overwhelm the drone with object detection and image processing, and worked with the Payload team to drop a water bottle and a beacon mid-flight using servos.
- Presented mapping and object detection improvement, image processing achievement, and correct GPU/CPU utilization to Lockheed Martin engineers during the Preliminary Design Review (PDR).
- Created ROS2 nodes handling the whole mission, including object detection (0.99 mAP@0.5), mapping, 4k image processing (under 0.7s each), and payload release over a target.

### Embedded Systems Member: Autonomous Ground Vehicle (University Club)

Feb 2026 - Jun 2026

Northrop Grumman Collaboration Project

Pomona, CA

- Presented telemetry connection between a drone and rover workflow to Northrop Grumman engineers, took their criticism and improved the software structure.
- Implemented Feature-driven development and a Lean approach to release new code every week and meet all the requirements before any demonstrations or meetings.
- Ran an authorized hardware review of a DS2431-class authenticator with a Bus Pirate v4 and logic analyzer, confirming a family-code mismatch and identifying clone silicon.
- Worked alongside the firmware team to develop bare-metal firmware for a pre-made microcontroller in C, directly manipulating registers (GPIO, timers, interrupts, etc.)

### Software Engineer: Autonomous Vehicle Lab (Research)

Apr. 2026 – Present

University Research Lab

Pomona, CA

- Built a closed-loop stepper controller (CL57T) on microcontroller with CAN telemetry over a TJA1051 transceiver to control a steering wheel through a computer, controller, or a joystick.
- Built CI/CD pipeline in GitHub Actions for the team to catch regression before merge, and created unit tests using GoogleTest and pytest to improve system reliability.
- Resolved priority-inversion bug by introducing mutex priority inheritance, restoring deterministic timing.
- Utilized AI agents to interface an MPU IMU over I2C on a Nucleo using CubeMX/HAL to stream 9-DOF telemetry.

## Technical Skills and Relevant Coursework

Languages and Tools: C, C++, Python, JavaScript, TypeScript, Verilog, VHDL, ARM Assembly, Bash, SQL, STM32 (bare-metal, register-level), ESP32, UART, SPI, I2C, DMA, interrupts, timers/PWM, logic analyzers, schematics, ROS2 (Humble/Iron), MAVLink, Gazebo, RPLIDAR, Jetson Orin Nano, YOLOv8/v11, PyTorch, TensorRT, ONNX, OpenCV, Roboflow, computer vision, mathematics, Google Cloud, deep learning, machine learning.

Linux, Git, GitHub Actions, Docker, Kubernetes, FastAPI, PostgreSQL, MongoDB, pytest, gtest.

Coursework: Microcontrollers & Embedded Systems, Circuit Analysis, Signals & Systems, Data Structures & Algorithms.

3 Languages: English, Russian, Armenian